EXPERIMENT 8

INPUT CODING

```
clc;
clear;
close all;

% Experiment 8: Time Response of a Continuous-Time System
% Task A: Underdamped Second-Order System

% Given specifications
wn = 5;        % Natural frequency in rad/s
zeta = 0.2;    % Damping ratio

% Transfer function: G(s) = wn^2 / (s^2 + 2*zeta*wn*s + wn^2)
num = [wn^2];
den = [1 2*zeta*wn wn^2];

G = tf(num, den);

% Display transfer function
disp('Transfer Function G(s):');
G

% Calculate and display poles
p = pole(G);
disp('Poles of the system:');
disp(p);

% Plot pole-zero map
figure;
pzmap(G);
grid on;
title('Pole-Zero Plot of Underdamped System');

% Plot unit step response
figure;
step(G);
grid on;
title('Step Response of Underdamped Second-Order System');
xlabel('Time (seconds)');
ylabel('Amplitude');

% Display step response characteristics
info = stepinfo(G);
disp('Step Response Characteristics:');
disp(info);
```

OUTPUT

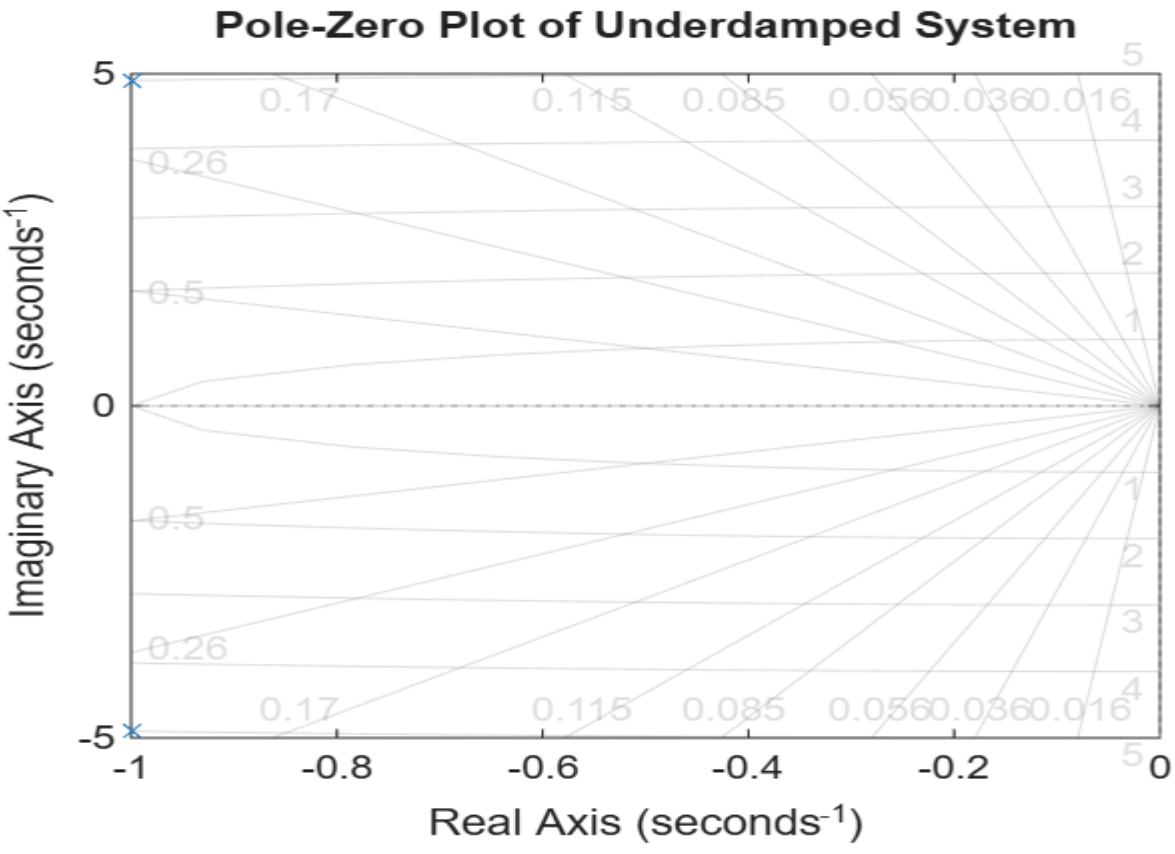
```
Transfer Function G(s):

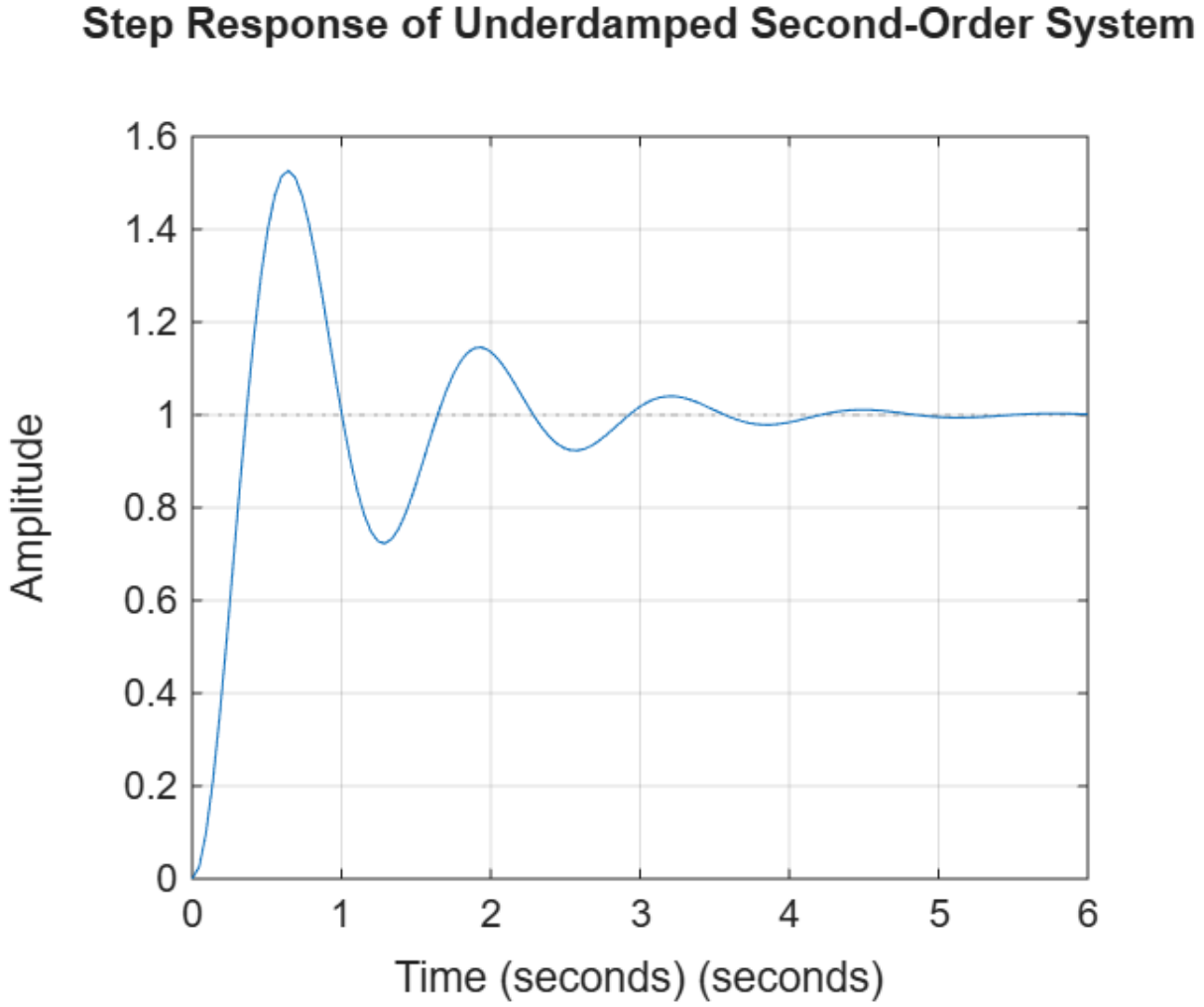
G =

        25
  --------------
  s^2 + 2 s + 25

Continuous-time transfer function.
Model Properties
Poles of the system:
  -1.0000 + 4.8990i
  -1.0000 - 4.8990i

Step Response Characteristics:
         RiseTime: 0.2411
    TransientTime: 3.9192
     SettlingTime: 3.9192
      SettlingMin: 0.7228
      SettlingMax: 1.5265
        Overshoot: 52.6542
       Undershoot: 0
             Peak: 1.5265
         PeakTime: 0.6447
```




Step Response of Underdamped Second-Order System
Amplitude
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
1
2
3
4
5
6
Time (seconds) (seconds)